Lec 9. Introduction to Parametric Tests

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Learning objectives

After this section, you should be able to ...

- understand the framework of hypothesis testing
- understand the method of significance test
- understand the parametric tests for common populations

9.1 Statistical hypothesis testing

- A **statistical hypothesis**, or just **hypothesis**(假设), is a claim or assertion either about the value of a single parameter (population characteristic or characteristic of a probability distribution), about the values of several parameters, or about the form of an entire probability distribution.

  Example hypotheses:

  - X is normally distributed
  - There is no value in using Vitamin D for preventing Covid
  - The defendant is innocent (innocent till proven guilty)

Fisher: we can simply measure the evidence for the null hypothesis.

Example

(The lady tasting tea) Dr. Muriel Bristol, a colleague of Fisher's, claimed that when drinking tea she could distinguish whether milk or tea was added to the cup first (she preferred milk first). To test her claim, Fisher asked her to taste eight cups of tea, four of which had milk added first and four of which had tea added first.

If she had no idea and guessed at random (H_0), n_{11} follows a hypergeometric distribution:

	Lady's Guess		
Poured first	Milk first	Tea first	total
Milk	$n_{11} = ?$		4
Tea			4
total	4	4	8

$$\mathrm{P}\left(n_{11} = i\right) = \binom{4}{i}\binom{4}{4-i}\Big/\binom{8}{4}, \quad i = 0, 1, 2, 3, 4$$

She actually got them all correct, which would have happened by chance with probability 0.014. Fisher concluded that since this probability was less that 0.05 that it was "statistically significant".

Statistical hypothesis testing

- The **null hypothesis**(原检验或零假设), denoted by H_0, is the claim that is initially assumed to be true (the "prior belief" claim).
- **A test of hypotheses** (假设检验) is a method for using sample data to decide whether the null hypothesis should be rejected.
- If a result is **statistically significant**(统计显著), that means it's unlikely to be explained solely by chance or random factors.
- The **alternative hypothesis**(对立假设或备择假设), denoted by H_1 or H_a, is the assertion that is contradictory to H_0

Neyman: we need alternative hypothesis: it helps us to choose a meaningful test statistic and design our study to have high power—a high chance of rejecting the null when the alternative is true. Without an alternative, we have no concept of power.

- The alternative hypothesis(对立检验或备择假设), H_1, is a claim about the population that is contradictory to H_0 and what we conclude when we reject H_0.

– H_1 : she had ability to distinguish. If $n_{11} = 3$, then under H_0,

$$\begin{aligned}\text{Evidence} &= \mathrm{P}\left(n_{11} \geq 3\right) = \mathrm{P}\left(n_{11} = 3\right) + \mathrm{P}\left(n_{11} = 4\right) \\ &= 0.229 + 0.014 = 0.243\end{aligned}$$

– H_1 : she guessed not at random. If $n_{11} = 3$, then under H_0,

$$\begin{aligned}\text{Evidence} &= \mathrm{P}\left(\text{no more likely than the observed } n_{11} = 3\right) \\ &= \frac{1}{70} + \frac{16}{70} + \frac{16}{70} + \frac{1}{70} = \frac{34}{70} \approx 0.486\end{aligned}$$

The Testing Framework

Let $X_1, \ldots, X_n$ be a sample from $f(x, \theta), \theta \in \Theta \subseteq \mathbb{R}$, consider a hypothesis about the parameter θ.

(1). **Null hypothesis H_0 and alternative hypothesis H_1**

$$H_0 : \theta \in \Theta_0 \leftrightarrow H_1 : \theta \in \Theta_1$$

where $\Theta_0 \cap \Theta_1 = \emptyset$. For $i = 0, 1$, if Θ_i consists of a single point, we call this a simple hypothesis. If Θ_i consists of more than one point, we call this a composite hypothesis.

Common testing problems:

	$H_0 \leftrightarrow H_1$
simple test	$\theta = \theta_0 \leftrightarrow \theta = \theta_1$
two-sided test	$\theta = \theta_0 \leftrightarrow \theta \neq \theta_0$
one-sided test	$\theta \leq \theta_0 \leftrightarrow \theta > \theta_1$
	$\theta \geq \theta_0 \leftrightarrow \theta < \theta_1$
	$\theta = \theta_0 \leftrightarrow \theta > \theta_1$
	$\theta = \theta_0 \leftrightarrow \theta < \theta_1$

The Testing Framework

The question is not whether H_0 is true or false. The question is whether there is sufficient evidence (in the sample) to reject H_0, much like a court case. Our possible actions are: reject H_0 or retain (don't reject) H_0.

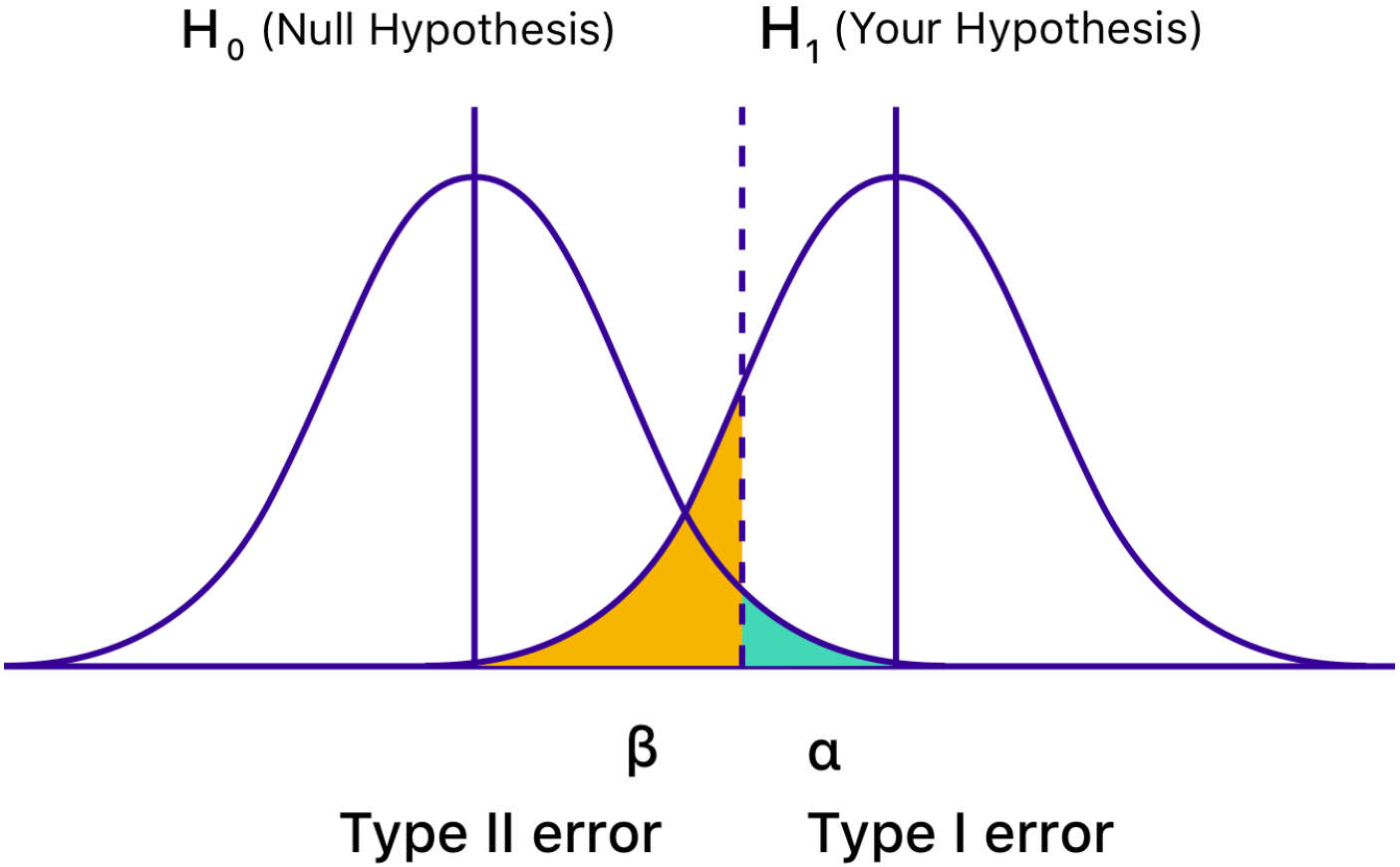
(2). **Type I and type II errors**

	Statistical Decision	
	Retain H_0	Reject H_0
H_0 true	$\surd$	$Type\ I\ error$ $(false\ positive)$
H_1 true	$Type\ II\ error$ $(false\ negative)$	$\surd$

– The risks of making these errors are **unavoidable** in hypothesis testing.

– $\alpha = P(Type\ I\ error),\ \beta = P(Type\ II\ error)$

e.g. $H_0 : \mu = \mu_0 \leftrightarrow H_1 : \mu = \mu_1$:



The Testing Framework

(3). **Testing statistic** $T = T(X_1, \ldots, X_n)$ is a function of the sample data on which the decision (reject H_0 or do not reject H_0) is to be based.

– A rejection region, R, for a test is defined by

$$R = \{(x_1, x_2, \ldots, x_n) \in \mathcal{X} \mid T(x_1, x_2, \ldots, x_n) \in A\},$$

where A is a subset of the real line. Usually, $R = \{T > c\}$, or $R = \{T < c\}$, c is called critical value.

– We define the test function ϕ_T by:

$$\phi_T(x_1, \ldots, x_n) = \begin{cases} 1 & \text{if } (x_1, \ldots, x_n) \in R \\ 0 & \text{otherwise} \end{cases}$$

We need to choose T and R so that the test has good statistical properties.

The Testing Framework

(4). **Power function** Suppose we reject H_0 when $(X_1, \ldots, X_n) \in R$. Define the power function of test ϕ_T by

$$\beta_{\phi_T}(\theta) = P_\theta(T \in A) = P_\theta\left((X_1, \ldots, X_n) \in R\right) = E_\theta \phi_T.$$

Then

$$P(Type\ I\ error) = \beta_{\phi_T}(\theta), \theta \in \Theta_0$$
$$P(Type\ II\ error) = 1 - \beta_{\phi_T}(\theta), \theta \in \Theta_1$$

- A test ϕ_T is (exact) size α (sometimes does not exist) if

$$\sup_{\theta \in \Theta_0} \beta_{\phi_T}(\theta) = \alpha.$$

- A test ϕ_T is level α (always exists) if

$$\sup_{\theta \in \Theta_0} \beta_{\phi_T}(\theta) \leq \alpha$$

We want $\beta_{\phi_T}(\theta)$ to be small when $\theta \in \Theta_0$ and we want $\beta_{\phi_T}(\theta)$ to be large when $\theta \in \Theta_1$.

The general strategies are: Given $\alpha \in [0, 1]$,

(a) find critical value such that $\beta_{\phi_T}(\theta) \leq \alpha$ for $\theta \in \Theta_0$, (i.e., a level α test, Fisher's idea); or

(b) maximize $\beta_{\phi_T}(\theta)$ for $\theta \in \Theta_1$ subject to (a) a level α UMP test, Neyman's idea

Unfortunately, type I and II errors cannot become arbitrarily small simultaneously at fixed sample size (Try the lab). Therefore, S(b) can be rarely achieved, it depends on the testing problem and sample distribution.

Steps for Tests of Significance

—the rejection region method

1. State the null hypotheses H_0 and the alternative hypothesis H_1 in terms of parameters, and decide on a significance level α

2. Specify a test statistic T(e.g., MLE), decide on the general form of the rejection region (with some unknown constants)

3. Let the test ϕ_T be level α test to determine unknown constants.

4. Decision: Reject H_0 if the sample falls in the rejection region and is otherwise not rejected.

5. State a conclusion in the context of the problem.

9.2 Tests about a Normal mean

Example

Let $X_1, \ldots, X_n$ be a sample from $N(\mu, \sigma^2)$, find the level α test for $H_0 : \mu = \mu_0$ with $H_a : \mu > \mu_0$, or $H_b : \mu < \mu_0$, or $H_c : \mu \neq \mu_0$.

Case 1: A Normal Population with Known σ

- $\bar{X}$ is the MLE of μ and $\bar{X} \sim N(\mu, \sigma^2/n)$
- Select test statistic $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma$, then by H_a the appropriate rejection region is $Z > c$
- Let $P(Z > c|H_0) \leq \alpha$, we have $c = z_\alpha$. That is

$$\phi_T(\mathbf{x}) = \begin{cases} 1 & z > z_\alpha \\ 0 & z \leq z_\alpha \end{cases}$$

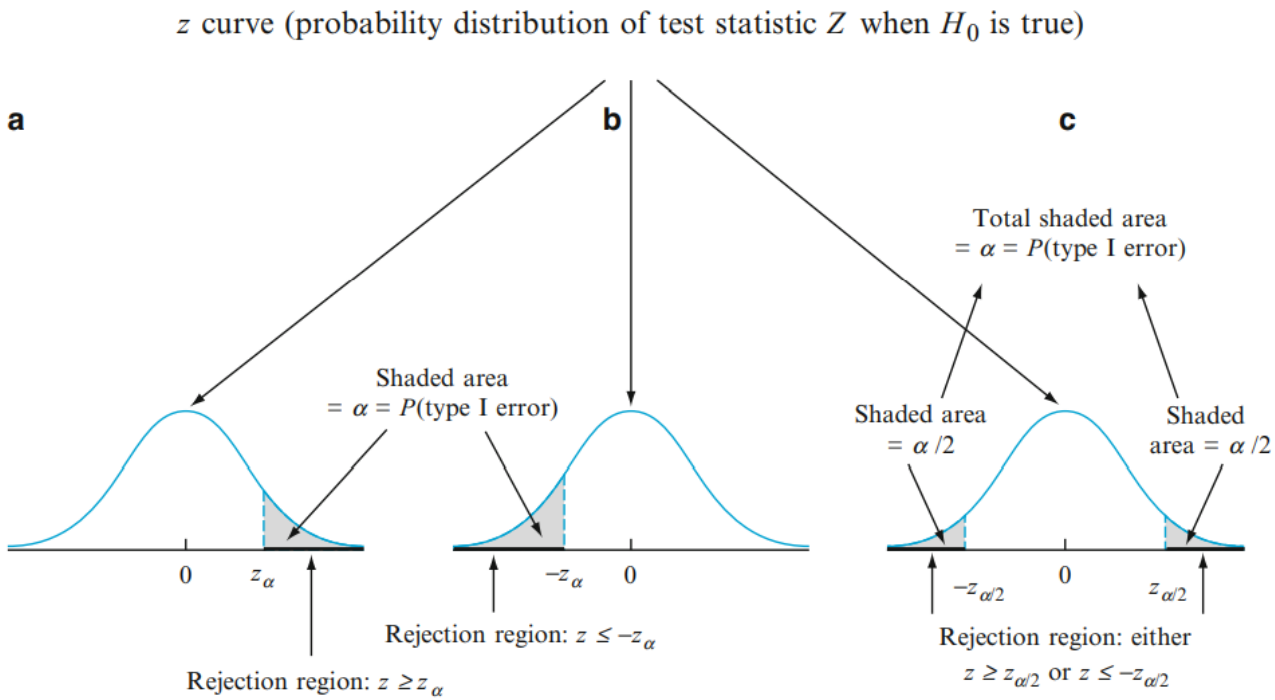
- State a conclusion

Summary: One-sample z Tests

Null hypothesis: $H_0 : \mu = \mu_0$

Test statistic value: $z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$, $N(0,1)$ when H_0 is true

Alternative Hypothesis	Rejection Region for Level α Test
$H_a : \mu > \mu_0$	$z \geq z_\alpha$ (upper-tailed test)
$H_b : \mu < \mu_0$	$z \leq -z_\alpha$ (lower-tailed test)
$H_c : \mu \neq \mu_0$	either $z \geq z_{\alpha/2}$ or $z \leq -z_{\alpha/2}$ (two-tailed test)



Case 2: A Normal Population with Unknown σ

Similar to z-test, but notice that $T = \frac{\bar{X}-\mu_0}{S/\sqrt{n}} \sim t_{n-1}$ when H_0 is true.

Summary: One-sample t Tests

Null hypothesis: $H_0 : \mu = \mu_0$

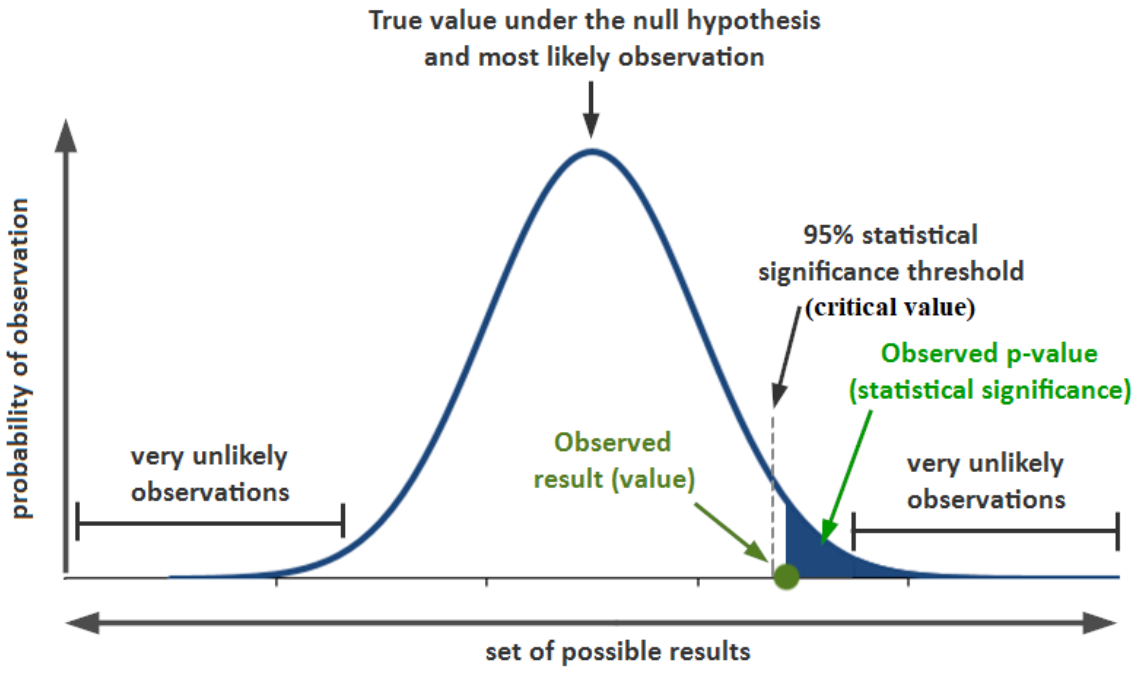
Test statistic value: $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$, t_{n-1} when H_0 is true

Alternative Hypothesis	Rejection Region for Level α Test
$H_{\mathrm{a}} : \mu > \mu_0$	$t \geq t_{n-1}(\alpha)$ (upper-tailed test)
$H_{\mathrm{b}} : \mu < \mu_0$	$t \leq -t_{n-1}(\alpha)$ (lower-tailed test)
$H_{\mathrm{c}} : \mu \neq \mu_0$	either $t \geq t_{n-1}(\alpha/2)$ or $t \leq -t_{n-1}(\alpha/2)$ (two-tailed t

9.3 P value

Definition

The $\boldsymbol{P}$-value is the probability of an observed (or more extreme) result assuming the null hypothesis is true.

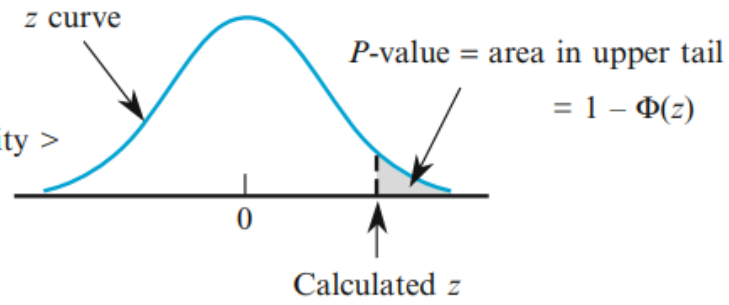


Proposition 1. *The P-value is the smallest significance level α at which the null hypothesis can be rejected. Because of this, the P-value is alternatively referred to as the observed significance level (OSL) for the data.*

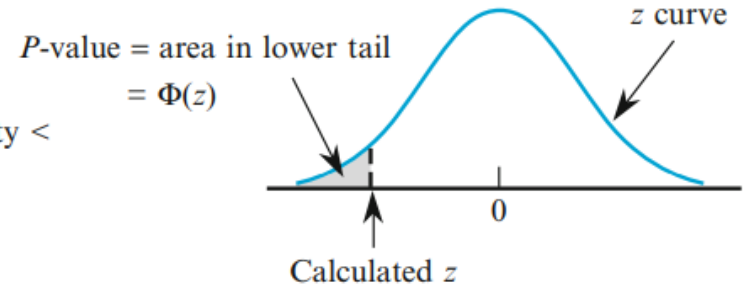
Decision rule based on p-value: Select a significance level α (as before, the desired type I error probability). Then reject H_0 if P-value $\leq \alpha$; do not reject H_0 if P-value $> \alpha$

P-Values for z Tests

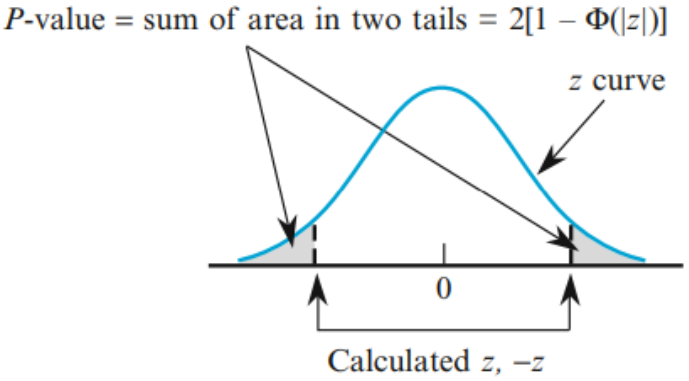
1. **Upper-tailed test**
   H_a contains the inequality >



2. **Lower-tailed test**
   H_a contains the inequality <



3. **Two-tailed test**
   H_a contains the inequality ≠



9.4 Statistical Versus Practical Significance

- A small P-value, which would ordinarily indicate **statistical significance** in that it would strongly suggest rejection of H_0 in favor of H_a, may be the result of a large sample size in combination with a departure from H_0 that has little **practical significance**.

  e.g., $H_0 : \mu = 100 \leftrightarrow H_a : \mu > 100$, $X \sim N(\mu, 100)$, true $\mu = 101$ would be a relatively inexpensive error, so we would not want this sample evidence to argue strongly for rejection of H_0 when $\bar{x} = 101$ is observed. But as sample size increases,

n	P_{value} when$\bar{x} = 101$	$\beta(101)$ for Level .01 Test
25	.3085	.9664
100	.1587	.9082
400	.0228	.6293
900	.0013	.2514
1600	.0000335	.0475
2500	.000000297	.0038
$10,000$	7.69×10^{-24}	.0000

9.5 The Duality between confidence intervals and tests

Recall the definition of $1-\alpha$ confidence region $S(\mathbf{X})$:

$$P_\theta\left(\theta \in S(\mathbf{X})\right) \geq 1-\alpha, \forall \theta \in \Theta$$

Next consider the testing framework where we test the hypothesis $H_0 : \theta = \theta_0$ for some specified value θ_0. Suppose we have a test $\phi(\mathbf{x}, \theta_0)$ with level α. Then the acceptance region

$$A\left(\theta_0\right) = \{\mathbf{x} : \phi(\mathbf{x}, \theta_0) = 0\}$$

is a subset of sample space $\mathcal{X}$ with probability at least $1-\alpha$. Then

Theorem 1 (Duality Theorem.). *Let $S(\mathbf{x}) = \{\theta_0 \in \Theta : \mathbf{x} \in A\left(\theta_0\right)\}$, the set of θ_0 being accepted under sample $\mathbf{x}$ and test ϕ, then*

$$P_{\theta_0}\left[\mathbf{X} \in A\left(\theta_0\right)\right] \geq 1-\alpha$$

if and only if $S(\mathbf{X})$ is a level $1-\alpha$ confidence region for θ.

Proof. For some specified θ_0, H_0 may be accepted, for other specified θ_0, H_0 may be rejected.

- Consider the set of θ_0 for which H_0 is accepted, that is, $S(\mathbf{x})$; By definition,

$$P_{\theta_0}(\theta_0 \in S(\mathbf{X})) = P_{\theta_0}(\mathbf{X} \in A(\theta_0)) \geq 1 - \alpha$$

  $S(\mathbf{X})$ is a level $1 - \alpha$ confidence region for θ.

- Conversely, if $S(\mathbf{X})$ is a level $1 - \alpha$ confidence region for θ, then define the test

$$\phi(\mathbf{x}, \theta_0) = \begin{cases} 1, & \theta_0 \notin S(\mathbf{x}) \\ 0, & \theta_0 \in S(\mathbf{x}) \end{cases}$$

  Then $\phi(\mathbf{X}, \theta_0)$ is a level α test for H_0.

$\square$